

PROJECT 5 — REVERSE-ENGINEERED CRYOGENIC INSTRUMENT HOUSING

Project Type: CAD / 3D Modelling Capstone

Client Brief

Reconstruct a cylindrical instrumentation housing from measurements, photographs and a partial 2D drawing where the original CAD data is unavailable. Build a robust parametric model, document assumptions, produce manufacturing drawings and validate exported CAD/mesh data.

Design Specification

- Cylindrical sensor/instrument housing
- Measured overall length and outer/inner diameters
- Measured wall thickness
- Measured flange and mounting-hole geometry
- Threaded interfaces and grooves where applicable
- Partial drawing plus photographs as source information

Parametric Requirements

OVERALL_LENGTH, OUTER_DIAMETER, INNER_DIAMETER, WALL_THICKNESS, FLANGE_DIAMETER, HOLE_PATTERN, THREAD_SIZE, GROOVE_DIMENSIONS, SENSOR_INSERTION_DEPTH and other measured functional dimensions.

Required Skills

- Reverse engineering
- Measurement interpretation
- Known vs measured vs inferred vs assumed data
- Master parameters / skeleton sketch
- Robust feature dependencies
- Threads and grooves
- Manufacturing drawings
- Section/detail views
- GD&T; where functionally justified
- STEP export/re-import
- STL tessellation and mesh QA

Client Approval Questions

- Which information is measured, known, inferred or assumed?
- What is the design intent?
- Which dimensions should be driving parameters?
- Which geometry should reference functional datums?
- What changes if sensor insertion depth increases by 5 mm?

- Which tolerances require explicit control?
- How do you prove the reconstructed model is geometrically and parametrically valid?

CAD QA / Validation

- Solid and topology validity
- Parametric regeneration test
- Feature dependency review
- Drawing/model consistency
- STEP round-trip validation
- STL watertight/manifold check
- Non-manifold edge / inverted-normal check
- Mesh density and tessellation comparison
- Dimensional comparison after export/re-import
- Manufacturing sanity check

Required Deliverables

- Native FreeCAD model (.FCStd)
- Manufacturing / engineering drawing (PDF)
- STEP export
- STL export where relevant
- Parameter/design-intent record
- CAD QA checklist/report